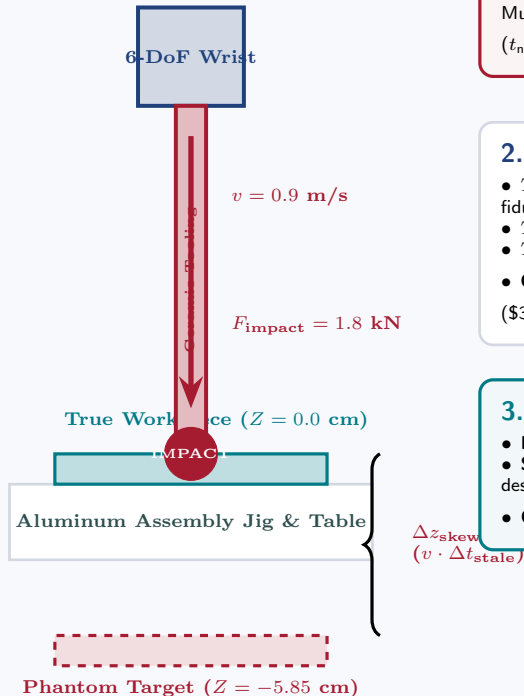


# ANATOMY OF A STALE FRAME TREE COLLISION

How multiplying asynchronous transforms creates a 5.85 cm vertical skew and a 1.8 kN tooling impact



## 1. Asynchronous Transform Multiplication

$$\mathbf{T}_{\text{target}}^{\text{base}}(t_{\text{now}}) = \mathbf{T}_{\text{cam}}^{\text{base}}(t_{\text{now}}) \cdot \mathbf{T}_{\text{target}}^{\text{cam}}(t_{\text{now}} - 65 \text{ ms})$$

Multiplying synchronous joint angles ( $t_{\text{now}}$ ) with stale camera coordinates ( $t_{\text{now}} - 65 \text{ ms}$ ) injects unmodeled vertical skew ( $\Delta z = 5.85 \text{ cm}$ ).

## 2. Microsecond Incident Chronology

- $T - 180 \text{ ms}$ : Forearm linkage rotates into optical path, occluding fiducials.
- $T - 65 \text{ ms}$ : Vision pipeline stalls; lookup returns cached transform.
- $T + 0 \text{ ms}$ : Tooling strikes fixture at  $0.9 \text{ m/s}$  ( $1.8 \text{ kN}$  impact force).
- **Consequence**: Ceramic tooling shatters; internal load cells damaged (\$3,200).

## 3. Physical AI Architectural Defenses

- **PTP-Indexed SRAM Ring Buffers**: Exact  $\text{se}(3)$  geodesic interpolation.
- **State TTL Leases** ( $\tau_{\text{ttl}} = 35 \text{ ms}$ ): Hard expiration halts downward descent.
- **Covariance Inflation**:  $\sigma_{\text{pos}}(t) = \sigma_0 + v \Delta t$  triggers Category 2 safe stop.